

Toward a Complete Digital Twin of Hub Sort Operations: An Idealized Multi-Layer Mathematical Framework Integrating External Logistics, Physical Infrastructure, and Human Factors

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Version: 2.1 — Empirical Grounding Extension (Five-Sort-Week Data, 05.04–05.08 2026)

Status: Empirically grounded extension of v2.0 — new sections calibrated against real operational data

Relation to v2.0: v2.1 preserves all v2.0 theory unchanged and adds §§16–21: CURE outbound loading, SEAS package composition, LIB service failure state, Zone 9-12 structural constant, weekly volume envelope, and empirical theorem status.

Relation to v1.4: v1.4 is a **restricted case** of this framework under the assumption that all external, infrastructure, and environmental layers are unobserved.

WARNING: This branch does not replace v1.4 or the incremental v1.5 path. It defines a long-horizon research target grounded in five-sort-week empirical data.

Abstract

The v1.4 framework establishes a mathematically rigorous three-timescale model of the UPS Chelmsford package sort, validated against 633 historical sorts. That framework treats the sort as a **closed system** — it ingests data only from iGate scanning and SOR staffing records, with no visibility into the external logistics network that determines the volume, timing, and composition of incoming freight.

This paper formulates **v2.0**, an idealized extension that opens the sort to its full causal environment. We introduce five additional observational layers: (1) an **external arrival layer** modeling feeder truck and delivery driver GPS positions as a marked point process with Bayesian ETA updating; (2) an **environmental layer** coupling weather state (temperature, precipitation, visibility, road conditions) to traffic delay amplification and accident probability via a coupled Markov-modulated process; (3) a **physical infrastructure layer** formalizing the conveyor belt system as a time-inhomogeneous M/G/1 queue with controlled service rate (belt speed), a jam probability model under high induction density, and chute accumulation dynamics; (4) a **package composition layer** connecting SLIC-code distributions and flowable/non-flowable fractions to belt capacity consumption and zone load imbalance; and (5) a **human factors layer** encoding fatigue degradation curves, experience heterogeneity, and break schedule discontinuities. These layers are unified in a **high-dimensional state space model** tracked by an Extended Kalman Filter, enabling predictive simulation of any observable quantity — PPH, belt congestion, staffing needs, or sort completion time — from pre-sort sensor data alone.

v2.0 is explicitly designated a **research vision branch**. It identifies what data must exist, what sensors must be deployed, and what mathematical infrastructure must be in place before this model can be realized in production. v1.4 remains the authoritative operational baseline; v1.5 will extend it incrementally. v2.0 is the north star.

Keywords: digital twin; package sorting; stochastic arrival process; M/G/1 queue; Extended Kalman Filter; weather-traffic coupling; fatigue model; induction rate control; GPS telematics; closed-loop optimization.

v2.1 Extension. This revision adds six new sections (§§16–21) grounded in real operational data from the Chelmsford Twilight sort week of 05.04–05.08, 2026 — five consecutive sorts totaling 531,148 packages across twelve PD belts. Section 16 integrates the CURE outbound trailer loading layer as the downstream counterpart to the v2.0 EKF’s inbound arrival model, introducing the Loading Bottleneck Theorem: above a utilization threshold of ~ 0.92 , increasing sort throughput on high-utilization lanes produces negative marginal utility on service quality. Section 17 grounds the package composition layer (§6) in SEAS quality data, establishing SMALLS SORT as the hub’s highest-leverage quality intervention point. Section 18 formalizes LIB events as the observable service failure state, deriving a Service Failure Rate metric and a structural LIB decay property with an empirical floor of ~ 50 – 60 HUB DERIVED events per Twilight sort. Section 19 establishes Zone 9-12 invariance: the four high-volume-destination belts maintain a structurally constant $31.1\% \pm 1.3\%$ share of total hub volume across all five sorts, enabling scalar-multiplication forecasting. Section 20 models the weekly volume gradient as a geometric decay with factor $\gamma_{approx} 0.938$, identifying a Wednesday anomaly consistent with the v3.6 staffing anomaly documented in the parallel research branch. Section 21 reports the empirical status of the original v2.0 theorems under this new data.

Additional Keywords: CURE trailer utilization; SEAS quality metrics; LIB service failure; smalls sort; zone share invariance; weekly volume envelope; loading bottleneck; empirical calibration.

1. Introduction

1.1 The Open-System Problem

Every quantity the v1.4 framework predicts — PPH trajectories, staffing needs, sort quality scores — is determined not only by what happens inside the building but by what is approaching it. A feeder delayed on I-495 by a winter storm delivers its 1,800 pieces twenty minutes late; the inbound ramp compresses, the induction rate spikes, jam probability increases, and sorters who would have been in steady-state are abruptly handling peak-phase density. The v1.4 model sees none of this until the packages are on the belt. It infers external dynamics from their internal consequences, after the fact.

This is not a failure of v1.4. It is an accurate description of what data is currently available. The Tracker and MasterTrendAnalysis operate on iGate and SOR exports — both are post-hoc records of what happened, not pre-hoc predictions of what is incoming. Under current data constraints, v1.4 is the correct model.

v2.0 asks: *if all relevant data existed and were accessible in real time, what would the complete mathematical model look like?*

1.2 Five New Causal Layers

The v1.4 framework is a **single-layer model**: it observes the interior of the sort (scans, hours, PPH) and predicts interior quantities. v2.0 adds five layers that feed causally into sort dynamics before they appear in iGate data:

Layer	Data Sources	Causal Role
External Arrival	Feeder GPS, delivery driver GPS, ORION manifests	Determines volume, timing, and phasing of package flow
Environmental	NOAA weather API, Waze/HERE traffic, DOT incident feeds	Modulates feeder ETAs, road delay, driver safety margins
Physical Infrastructure	Belt speed sensors, induction rate counters, chute load cells	Sets throughput ceiling, jam probability, sort capacity
Package Composition	SLIC manifests, package dimensions (vision system), weight sensors	Determines belt capacity consumption, zone load distribution
Human Factors	Biometric wearables (opt-in), time-on-task, break records	Modulates effective PPH as function of fatigue and experience

1.3 v2.0 as Idealized Restriction of a General Model

Formally, the v1.4 framework is obtained from v2.0 by conditioning on the unobservability of all five new layers. Every theorem in v1.4 remains valid in v2.0 — they are simply theorems about the marginal dynamics of the iGate/SOR subsystem when external state is integrated out. v2.0 does not revise v1.4; it embeds it.

1.4 Organization

Section 2 defines the complete system architecture. Sections 3–7 develop each new layer mathematically. Section 8 unifies them in a high-dimensional state space model with Extended Kalman Filter. Section 9 derives the integrated optimization framework. Section 10 presents the digital twin architecture. Section 11 formalizes the connection to v1.4 as a restricted case. Section 12 enumerates the data requirements that must be met before v2.0 can be instantiated. Sections 13–14 address limitations and future research directions. (*v2.1 additions*) Sections 16–21 provide empirical grounding from the five-sort week of 05.04–05.08, 2026: CURE outbound loading integration (§16), SEAS package composition calibration (§17), LIB as service failure state (§18), Zone 9-12 structural constant (§19), weekly volume envelope model (§20), and empirical theorem status (§21).

2. System Architecture

2.1 The Hub as a Controlled Queueing Network

The UPS Chelmsford sort can be modeled as a **controlled queueing network** $\mathcal{H} = (\mathcal{N}, \mathcal{E}, \mathcal{C}, \mathcal{U})$ where:

- $\mathcal{N}$ is the set of nodes: induction stations, belt segments, sort chutes, unload doors, load doors
- $\mathcal{E}$ is the set of directed edges (package flow paths)

- $\mathcal{C} : \mathcal{E} \rightarrow \mathbb{R}_+$ assigns capacity (packages/min) to each edge
- $\mathcal{U}$ is the control space: belt speed settings, induction rate targets, staffing assignments, door open/close states

Packages arrive at induction nodes according to an externally-driven **arrival process** $\Lambda(t)$ (Section 3), traverse the belt network subject to capacity constraints $\mathcal{C}$ and jam dynamics (Section 5), and exit at sort chutes with a routing distribution determined by the sorter channel model $\mathbf{M}$ (v1.4, §4).

2.2 Multi-Layer State Vector

The complete state of the system at time $t \in [0, 240]$ (minutes from sort start) is:

$$\mathbf{X}(t) = [\underbrace{\mathbf{x}_{\text{ext}}(t)}_{\text{External}}, \underbrace{\mathbf{x}_{\text{env}}(t)}_{\text{Environmental}}, \underbrace{\mathbf{x}_{\text{infra}}(t)}_{\text{Infrastructure}}, \underbrace{\mathbf{x}_{\text{pkg}}(t)}_{\text{Package}}, \underbrace{\mathbf{x}_{\text{human}}(t)}_{\text{Human}}, \underbrace{\mathbf{x}_{\text{ops}}(t)}_{\text{v1.4 Ops}}]$$

The v1.4 state vector $\mathbf{x}_{\text{ops}}(t)$ — containing PPH, staffing counts, fidelity score, SQS components — is the **observable projection** of the full state $\mathbf{X}(t)$ onto the iGate/SOR subspace. All upstream layers are latent in the v1.4 model but explicit in v2.0.

3. External Arrival Layer

3.1 The Freight Arrival Process

Definition 3.1 (Marked Point Process). The inbound freight arrival process is a marked point process $\Phi = \{(\tau_i, \mathbf{m}_i)\}_{i=1}^N$ on $[t_0 - \Delta, T]$ (beginning Δ minutes before sort start $t_0 = 0$, ending at sort close $T = 240$), where:

- $\tau_i \in \mathbb{R}$ is the arrival time of vehicle i at the hub dock
- $\mathbf{m}_i = (V_i, \mathbf{s}_i, \text{type}_i)$ is the mark: total piece count V_i , SLIC vector $\mathbf{s}_i \in \mathbb{R}^{|\mathcal{Z}|}$ (zone load distribution), and vehicle type $\text{type}_i \in \{\text{feeder}, \text{P-car}, \text{air}\}$

The arrival intensity function $\lambda(t)$ is non-homogeneous, with characteristic feeder pulses at approximately $t \in \{-60, 0, 30, 90\}$ minutes relative to sort start (pre-loaded, first wave, second wave, peak feeder wave) and a distributed P-car tail through $t \in [30, 180]$ as delivery drivers return from route.

Definition 3.2 (GPS-Based ETA Estimator). For each vehicle i with GPS position $\mathbf{p}_i(t) \in \mathbb{R}^2$, velocity $v_i(t)$, and route $\mathcal{R}_i$ (polyline to hub), the ETA estimate at time t is:

$$\hat{\tau}_i(t) = t + \int_{\mathbf{p}_i(t)}^{\mathbf{p}_{\text{hub}}} \frac{d\ell}{v_i(\ell, t, W(t))} + \epsilon_i(t)$$

where $v_i(\ell, t, W(t))$ is speed at road segment ℓ conditioned on weather state $W(t)$ (Section 4), and $\epsilon_i(t) \sim \mathcal{N}(0, \sigma_{\tau,i}^2(t))$ captures GPS uncertainty and unmodeled delays. ETA uncertainty $\sigma_{\tau,i}^2(t)$ decreases as i approaches: $\sigma_{\tau,i}^2(t) = \sigma_0^2 \cdot d_i(t)/d_0$ where $d_i(t)$ is remaining distance.

Theorem 3.1 (Volume Arrival Forecast). The expected cumulative volume arriving by time t is:

$$\mathbb{E}[V_{\text{total}}(t)] = \sum_{i=1}^N V_i \cdot \Phi\left(\frac{t - \hat{\tau}_i(t_0)}{\sigma_{\tau,i}(t_0)}\right)$$

where $\Phi(\cdot)$ is the standard normal CDF. This provides a probabilistic inbound forecast from the pre-sort GPS snapshot, with confidence bands narrowing as arrival time approaches.

Proof. Volume from vehicle i arrives at hub at time τ_i . Given $\tau_i \sim \mathcal{N}(\hat{\tau}_i, \sigma_{\tau,i}^2)$, $P(\tau_i \leq t) = \Phi((t - \hat{\tau}_i)/\sigma_{\tau,i})$. Total expected volume is by linearity of expectation over independent vehicle arrivals. $\square$

3.2 Feeder vs. Package Car Arrival Dynamics

Definition 3.3 (Feeder Pulse Model). A feeder arrival at time τ_i generates a **volume pulse** at the induction nodes:

$$\lambda_{\text{induct}}(t|\tau_i, V_i) = V_i \cdot \frac{r_{\text{unload}}}{\Delta t_{\text{unload}}} \cdot \mathbf{1}[\tau_i \leq t \leq \tau_i + \Delta t_{\text{unload}}]$$

where r_{unload} is the unload rate (pieces/min, dependent on staffing at that door), and $\Delta t_{\text{unload}} = V_i/r_{\text{unload}}$ is the unload duration. Multiple simultaneous feeders create **superimposed pulses** — a key driver of induction rate spikes and downstream belt congestion.

Definition 3.4 (P-Car Return Distribution). Delivery driver (P-car) return times follow a distribution shaped by route density and completion rate:

$$f_{\text{Pcar}}(\tau) = \text{LogNormal}(\mu_{\text{route}}, \sigma_{\text{route}}^2) \cdot \mathbf{1}[\tau \in [30, 180]]$$

with parameters estimated from historical ORION route completion data. Each P-car contributes a small volume ($\bar{V}_{\text{Pcar}} \approx 5\text{--}30$ pieces: undeliverables, pickups, returns), but the aggregate over 80–120 simultaneous drivers creates a sustained background arrival rate.

3.3 Volume Forecast Integration with Kalman Filter

The GPS-based volume forecast (Theorem 3.1) is fed as an **exogenous input** to the v1.4 Kalman filter prediction step:

$$\text{PPH}_{\text{predicted}}(t + \delta) = f\left(\text{PPH}(t), \left.\frac{d\mathbb{E}[V_{\text{total}}]}{dt}\right|_t, N(t), \phi(t)\right)$$

where $\phi(t) \in \{\text{Ramp}, \text{Steady}, \text{Wind-Down}, \text{Post}\}$ is the current phase and $d\mathbb{E}[V_{\text{total}}]/dt$ is the expected induction rate implied by approaching vehicles.

4. Environmental Layer

4.1 Weather State Markov Chain

Definition 4.1 (Weather State Space). The weather state $W(t)$ takes values in $\mathcal{W} = \{\text{Clear, Rain, HeavyRain, Snow, Ice, Fog}\}$ and evolves as a continuous-time Markov chain with generator matrix $\mathbf{Q}_W$ estimated from NOAA historical data for the Chelmsford, MA area by month and season.

Definition 4.2 (Weather-Speed Coupling). Road segment speed under weather state w is:

$$v(\ell, w) = v_{\text{free}}(\ell) \cdot \alpha_w \cdot \beta(\ell, w)$$

where $v_{\text{free}}(\ell)$ is free-flow speed, $\alpha_w \in (0, 1]$ is a weather attenuation factor:

State w	α_w (highway)	α_w (local)
Clear	1.00	1.00
Rain	0.85	0.80
HeavyRain	0.70	0.65
Snow	0.55	0.45
Ice	0.35	0.25
Fog	0.75	0.70

and $\beta(\ell, w)$ is a road-type correction (surface treatment, grade, curvature) estimated from DOT road geometry data.

4.2 Traffic Delay Amplification

Definition 4.3 (Traffic Density State). Traffic density on route $\mathcal{R}_i$ at time t is modeled via the Bureau of Public Roads (BPR) function:

$$v_{\text{traffic}}(\ell, t) = \frac{v_{\text{free}}(\ell)}{1 + 0.15 \left(\frac{q(\ell, t)}{c(\ell)} \right)^4}$$

where $q(\ell, t)$ is observed traffic flow (vehicles/hr) from HERE or Waze API, and $c(\ell)$ is road capacity (vehicles/hr/lane $\times$ lanes). Weather and traffic effects compound multiplicatively:

$$v_{\text{effective}}(\ell, t, w) = v_{\text{traffic}}(\ell, t) \cdot \alpha_w \cdot \beta(\ell, w)$$

Definition 4.4 (Accident Probability Model). The probability of a traffic incident on route $\mathcal{R}_i$ during the sort window is:

$$P(\text{accident on } \mathcal{R}_i) = 1 - \exp \left(- \int_{\mathcal{R}_i} \lambda_{\text{acc}}(\ell, w, q) d\ell \right)$$

where $\lambda_{\text{acc}}(\ell, w, q)$ is the local accident rate per unit length, modeled as:

$$\lambda_{\text{acc}}(\ell, w, q) = \lambda_0(\ell) \cdot \gamma_w \cdot \left(\frac{q(\ell)}{c(\ell)} \right)^\delta$$

with $\lambda_0(\ell)$ from historical MASSDOT incident data, γ_w a weather multiplier (Ice: 4.2 $\times$, Snow: 2.8 $\times$, HeavyRain: 1.6 $\times$, Rain: 1.2 $\times$, Clear: 1.0 $\times$), and $\delta \approx 1.4$ from empirical accident-density correlation studies (Lord & Mannering, 2010).

Theorem 4.1 (Delay Cascade). An accident at road segment ℓ^* with clearance time T_{clear} induces a delay cascade on all vehicles routed through ℓ^* with expected arrival delay:

$$\mathbb{E}[\Delta\tau_i | \text{accident at } \ell^*, T_{\text{clear}}] = T_{\text{clear}} + \frac{L_{\text{queue}}(t^*)}{v_{\text{discharge}}}$$

where $L_{\text{queue}}(t^*)$ is the queue length at accident clearance and $v_{\text{discharge}}$ is the discharge rate. This cascade propagates through the GPS-ETA estimator (Definition 3.2), updating $\hat{\tau}_i$ and triggering re-optimization of the staffing plan (Section 9.2).

4.3 Temperature Effects on Operations

Beyond traffic, weather directly affects sort operations:

Definition 4.5 (Heat-Fatigue Coupling). Sorter effective PPH under high ambient temperature T_{amb} ($^{\circ}\text{F}$) degrades according to:

$$\text{PPH}_{\text{eff}}(t, T_{\text{amb}}) = \text{PPH}_{\text{nominal}}(t) \cdot (1 - \kappa_T \cdot \max(0, T_{\text{amb}} - T_{\text{threshold}}))^+$$

with threshold $T_{\text{threshold}} \approx 80^{\circ}\text{F}$ (inside the building, affected by ambient outdoor temperature and building HVAC), and κ_T estimated from productivity studies in warm-environment manual labor (Seppänen et al., 2006: $\kappa_T \approx 0.02/^{\circ}\text{F}$ above threshold for sustained physical work). Cold temperatures ($T_{\text{amb}} < 50^{\circ}\text{F}$) produce slower initial ramp but may preserve late-sort PPH.

Definition 4.6 (Ice/Snow Loading Effect). Packages arriving via feeders during heavy precipitation carry surface moisture, which increases conveyor belt friction variability and can elevate jam probability (Section 5.3) by a factor $\eta_{\text{ice}} \in [1.1, 1.4]$ during Ice/Snow weather states.

5. Physical Infrastructure Layer

5.1 The Conveyor Belt as a Queue

Definition 5.1 (Belt Queue Model). Each belt segment $b \in \mathcal{B}$ is modeled as a time-inhomogeneous M/G/1 queue with:

- **Arrival rate:** $\lambda_b(t)$ — packages per minute arriving at segment b from upstream
- **Service rate:** $\mu_b(s_b) = s_b \cdot \rho_b$ — packages per minute processed, where s_b is belt speed (m/min) and ρ_b is packages per meter of belt (dependent on package size distribution)
- **Traffic intensity:** $\rho_b(t) = \lambda_b(t)/\mu_b(s_b)$

For stable belt flow, $\rho_b(t) < 1$ for all b and all t . The M/G/1 Pollaczek-Khinchine formula gives expected queue length:

$$\mathbb{E}[L_b] = \rho_b + \frac{\rho_b^2 + \lambda_b^2 \text{Var}[S_b]}{2(1 - \rho_b)}$$

where S_b is the service time distribution (dependent on package size variance — see §6.1).

Definition 5.2 (Belt Speed Control). Belt speed $s_b(t) \in [s_{\min}, s_{\max}]$ is a **control variable**. Increasing s_b raises μ_b and reduces congestion, but:

1. Increases package impact velocity at chute deflectors, elevating missort and damage risk
2. Reduces sorter reaction time per package, reducing effective scan accuracy (sorter DMC model)
3. Exceeds safe operating parameters above $s_{\max}$

The belt speed control policy (Section 9.3) optimizes throughput subject to these constraints.

5.2 Induction Rate Dynamics

Definition 5.3 (Induction Rate). The induction rate $\lambda_{\text{induct}}(t)$ (packages/min entering the system) is determined by unload staffing, feeder availability, and physical induction station capacity:

$$\lambda_{\text{induct}}(t) = \min(\lambda_{\text{raw}}(t), C_{\text{induct}}, \mu_{\text{main}}(s_{\text{main}}))$$

where $\lambda_{\text{raw}}(t) = \sum_{i: \tau_i \leq t} r_{\text{unload}, i}(t)$ is the raw arrival rate from active unload doors, C_{induct} is the physical induction station capacity (a fixed infrastructure constant), and $\mu_{\text{main}}(s_{\text{main}})$ is the main belt throughput capacity. The minimum enforces the bottleneck constraint.

Theorem 5.1 (Induction Ceiling). When $\lambda_{\text{raw}}(t) > \min(C_{\text{induct}}, \mu_{\text{main}})$, packages accumulate at the induction buffer at rate:

$$\frac{dB_{\text{induct}}}{dt} = \lambda_{\text{raw}}(t) - \min(C_{\text{induct}}, \mu_{\text{main}})$$

The buffer B_{induct} clears only when $\lambda_{\text{raw}}(t)$ drops below capacity. Buffer overflow forces door closure and package staging on the dock floor — a directly observable operational disruption. The total sort duration extends by $\Delta T_{\text{buffer}} = B_{\text{peak}}/(\mu_{\text{main}} - \bar{\lambda}_{\text{tail}})$ where $\bar{\lambda}_{\text{tail}}$ is average arrival rate during wind-down.

5.3 Jam Probability Model

A belt jam occurs when packages accumulate in a segment faster than the segment can process them, creating a physical blockage. Jams have two primary drivers: (a) belt density exceeding safe separation, and (b) irregular packages (oversized, wet, damaged) bridging adjacent packages.

Definition 5.4 (Jam Hazard Function). The instantaneous jam hazard rate at segment b is:

$$h_b(t) = h_0 \cdot \exp(\alpha_\rho(\rho_b(t) - \rho^*) + \alpha_{\text{irr}}f_{\text{irr}}(t) + \alpha_{\text{ice}}\eta_{\text{ice}}(t))$$

where h_0 is the baseline jam rate (calibrated per-belt from historical maintenance logs), $\rho^* \approx 0.75$ is the critical density above which jam risk increases sharply, $f_{\text{irr}}(t)$ is the fraction of irregular packages in the current induction stream, and η_{ice} is the weather loading factor (Definition 4.6).

Definition 5.5 (Jam Recovery Model). A jam at segment b stops all flow through b for duration $T_{\text{jam}}^{(b)} \sim \text{Weibull}(k_b, \lambda_{\text{jam}}^{(b)})$, calibrated from historical jam response times. During jam recovery:

- Upstream segments accumulate packages at their input rate
- Downstream zones lose induction volume: $V_{\text{gap}} = \text{PPH}_{\text{design}} \cdot T_{\text{jam}}^{(b)}$ (v1.4 §13, Layer 2 distortion)
- Jam-breaker employees are reassigned, triggering the v1.4 overhead reclassification effect

This formalizes the **connection between jam dynamics and the v1.4 jam-breaker distortion model**: the Layer 1 and Layer 2 distortions in v1.4 §13 are the observable iGate/SOR consequences of a jam event whose physical dynamics are modeled here.

5.4 Chute Capacity and Sort Destination Accumulation

Definition 5.6 (Chute Queue). Each sort chute $c \in \mathcal{C}$ has physical capacity K_c (packages) and receives packages at rate $\mu_c(t)$ determined by the routing distribution and belt flow. Packages are removed at rate $r_c(t)$ (loading to outbound trailer). The chute occupancy evolves as:

$$\frac{dQ_c}{dt} = \mu_c(t) - r_c(t), \quad Q_c \in [0, K_c]$$

When $Q_c = K_c$ (chute full), the belt must hold packages at that deflector — creating a **back-pressure wave** that propagates upstream, reducing effective belt throughput. Chute fullness is a leading indicator of sort completion and outbound trailer load progress.

6. Package Composition Layer

6.1 Package Size Distribution and Belt Capacity

Definition 6.1 (Package Size Distribution). Packages are characterized by their footprint area a_i (cm²) on the belt. The empirical distribution F_a across a sort can be approximated as a mixture of two log-normals:

$$F_a = \pi_{\text{small}} \cdot \text{LogN}(\mu_s, \sigma_s^2) + (1 - \pi_{\text{small}}) \cdot \text{LogN}(\mu_l, \sigma_l^2)$$

corresponding to small/soft-goods packages (envelopes, polybags: $\mu_s \approx 200$ cm²) and large/box packages ($\mu_l \approx 800$ cm²). The mixing weight π_{small} varies by day-of-week and retail season.

Definition 6.2 (Belt Density). Effective belt density (packages/meter) is:

$$\rho_b = \frac{\bar{a}}{w_b \cdot g_{\min}}$$

where $\bar{a} = \mathbb{E}[a_i]$ is mean package footprint, w_b is belt width, and $g_{\min}$ is minimum required separation gap. Higher $\bar{a}$ (larger packages) reduces ρ_b and lowers the effective throughput capacity — a direct link between package composition and belt physics.

6.2 Flowable vs. Non-Flowable Classification

Definition 6.3 (Flowability). A package is **flowable** if it can be inducted through the standard chute deflector at belt speed s_b without requiring manual assist. Let $p_{\text{flow}}(t)$ be the fraction of flowable packages in the current induction stream. Non-flowable packages require manual deflection — reducing effective sorter PPH during high non-flowable periods (the “cube surge” effect observed operationally).

Theorem 6.1 (Effective PPH under Flowability Mix). Under fraction p_{flow} of flowable packages and $1 - p_{\text{flow}}$ non-flowable:

$$\text{PPH}_{\text{eff}} = p_{\text{flow}} \cdot \text{PPH}_{\text{flow}} + (1 - p_{\text{flow}}) \cdot \text{PPH}_{\text{nf}}$$

where $\text{PPH}_{\text{flow}} > \text{PPH}_{\text{nf}}$ reflects the additional time per non-flowable package. In steady-state, if non-flowable fraction rises sharply (e.g., a bulk-goods feeder arrives mid-sort), the sort will register a transient PPH drop without any change in staffing or effort — a composition effect that cannot be identified from iGate data alone.

6.3 SLIC-to-Zone Load Distribution

Definition 6.4 (SLIC Load Vector). Each feeder’s manifest carries a SLIC vector $\mathbf{s}_i = (s_{iz})_{z \in \mathcal{Z}} \in \mathbb{R}_+^{|\mathcal{Z}|}$ where s_{iz} is the number of packages destined for zone z . The aggregate zone load at time t is:

$$\mathbf{L}(t) = \sum_{i: \tau_i \leq t} \text{fraction_processed}_i(t) \cdot \mathbf{s}_i$$

Load imbalance $\sigma_L(t) = \text{std}(\mathbf{L}(t))$ is a measure of zone unbalance. High σ_L implies that some zones are overloaded (chutes filling faster, staffing understressed elsewhere) while others are underloaded — motivating the borrow/loan optimizer in v1.4 §14.

Theorem 6.2 (Pre-Sort Zone Load Forecast). Given GPS-derived feeder ETA estimates $\hat{\tau}_i$ and ORION manifests $\mathbf{s}_i$, the expected zone load by sort end is forecastable before the first package is inducted:

$$\hat{\mathbf{L}}_{\text{final}} = \sum_{i=1}^N \mathbf{s}_i \cdot P(\tau_i < T)$$

This forecast enables **pre-sort staffing optimization**: set initial zone allocations proportional to $\hat{\mathbf{L}}_{z, \text{final}}$ rather than waiting for iGate data to reveal imbalance reactively.

7. Human Factors Layer

7.1 Fatigue Degradation Curves

Physical labor productivity is not constant over a 4-hour sort. Empirical studies in manual materials handling (Garg et al., 1978; NIOSH, 2021) establish that sustained high-intensity sorting leads to measurable performance degradation.

Definition 7.1 (Fatigue Factor). The fatigue factor for sorter j at time t into the sort is:

$$F_j(t) = 1 - \frac{\kappa_F \cdot t^{\gamma_F}}{1 + \kappa_F \cdot t^{\gamma_F}}$$

with $\kappa_F > 0$ and $\gamma_F \in (0.5, 1.5]$ calibrated from physiological workload studies. $F_j(0) = 1$ (no fatigue), $F_j(t) \rightarrow (1 - \kappa_F)/(1 + \kappa_F)$ as $t \rightarrow \infty$ (saturation). The exponent γ_F controls the onset speed: low γ_F produces early-sort degradation, high γ_F produces late-sort cliff.

Definition 7.2 (Experience-Modulated PPH). The expected PPH for sorter j with experience e_j (sorts completed) at time t is:

$$\text{PPH}_j(t) = \text{PPH}_j^\infty \cdot (1 - \exp(-\theta e_j)) \cdot F_j(t) \cdot (1 - \kappa_T \Delta T_{\text{amb}})^+$$

where PPH_j^∞ is the asymptotic skill level (estimated via the v1.4 hierarchical Beta-Bernoulli model), θ is the learning rate (experience curve parameter), and ΔT_{amb} is the temperature excess from Definition 4.5.

7.2 Break Schedule Discontinuities

Definition 7.3 (Break Event). A scheduled break for sorter j at time $t_{\text{break}}^{(j)}$ of duration $\Delta_b^{(j)}$ induces a discontinuity in the staffing count $N(t)$ modeled in v1.4 as a jump process. In v2.0, breaks are **pre-scheduled events** in the state space:

$$N(t) = N_0 - \sum_j \mathbf{1} \left[t \in [t_{\text{break}}^{(j)}, t_{\text{break}}^{(j)} + \Delta_b^{(j)}] \right]$$

The break schedule is a known control input, enabling the Kalman filter to treat breaks as **known interventions** rather than unexplained staffing drops — improving prediction accuracy during the typically-degraded Wind-Down phase.

7.3 Sorter Heterogeneity and Assignment Optimization

Definition 7.4 (Skill-Weighted Staffing). Given per-sorter skill estimates $\hat{\theta}_j^{(z)}$ for belt family z (from the v1.4 hierarchical Bayesian model), the expected PPH for assignment $\sigma : j \mapsto z$ is:

$$\text{PPH}_{\text{assignment}}(\sigma) = \sum_j \hat{\text{PPH}}_j^{(\sigma(j))} \cdot F_j(t)$$

The optimal assignment $\sigma^*(t)$ maximizes $\text{PPH}_{\text{assignment}}(\sigma)$ subject to zone minimum staffing constraints. This is a weighted bipartite matching problem solvable in $O(n^3)$ via the Hungarian algorithm, or approximately in $O(n^2)$ via greedy skill-ranking.

8. Unified State Space Model

8.1 Full State Vector

The complete v2.0 state vector at time t is:

$$\mathbf{X}(t) = \begin{bmatrix} \hat{\tau} & \text{(feeder ETA vector, } N_F\text{-dim)} \\ W(t) & \text{(weather state, categorical)} \\ \lambda_b(t) & \text{(per-belt arrival rates, } |\mathcal{B}|\text{-dim)} \\ \rho_b(t) & \text{(per-belt occupancy, } |\mathcal{B}|\text{-dim)} \\ \mathbf{Q}_c(t) & \text{(chute queues, } |\mathcal{C}|\text{-dim)} \\ p_{\text{flow}}(t) & \text{(flowable fraction)} \\ \mathbf{L}(t) & \text{(zone load vector, } |\mathcal{Z}|\text{-dim)} \\ \mathbf{F}(t) & \text{(sorter fatigue vector, } N(t)\text{-dim)} \\ \mathbf{x}_{\text{ops}}(t) & \text{(v1.4 ops state: PPH, SQS, fidelity, } N(t)) \end{bmatrix}$$

Total state dimension: $\approx 2N_F + 2|\mathcal{B}| + |\mathcal{C}| + 2|\mathcal{Z}| + N(t) + 6$. For Chelmsford parameters ($N_F \approx 20$ feeders, $|\mathcal{B}| \approx 30$ belt segments, $|\mathcal{C}| \approx 120$ chutes, $|\mathcal{Z}| \approx 8$ zones, $N(t) \approx 40$ sorters), the state dimension is approximately **300**.

8.2 Extended Kalman Filter

Because the system dynamics are nonlinear (M/G/1 queue congestion, fatigue curves, jam hazard), a standard linear Kalman filter is insufficient. The **Extended Kalman Filter (EKF)** linearizes the dynamics around the current state estimate.

Definition 8.1 (EKF Prediction Step). At each snapshot interval $\delta t = 15$ min, the state prediction is:

$$\begin{aligned} \hat{\mathbf{X}}(t + \delta t|t) &= \mathbf{f}(\hat{\mathbf{X}}(t|t), \mathbf{u}(t)) \\ \mathbf{P}(t + \delta t|t) &= \mathbf{F}_t \mathbf{P}(t|t) \mathbf{F}_t^\top + \mathbf{Q}_t \end{aligned}$$

where $\mathbf{f}(\cdot)$ is the nonlinear state transition (incorporating queue dynamics, fatigue evolution, weather Markov transitions), $\mathbf{u}(t)$ is the control input (belt speed, staffing assignment), $\mathbf{F}_t = \partial \mathbf{f} / \partial \mathbf{X}|_{\hat{\mathbf{X}}(t|t)}$ is the Jacobian, and $\mathbf{Q}_t$ is process noise (modeling unmodeled disturbances).

Definition 8.2 (EKF Update Step). When a new iGate/SOR snapshot $\mathbf{z}(t)$ arrives:

$$\begin{aligned} \mathbf{K}(t) &= \mathbf{P}(t|t-1) \mathbf{H}_t^\top (\mathbf{H}_t \mathbf{P}(t|t-1) \mathbf{H}_t^\top + \mathbf{R}(t))^{-1} \\ \hat{\mathbf{X}}(t|t) &= \hat{\mathbf{X}}(t|t-1) + \mathbf{K}(t) (\mathbf{z}(t) - \mathbf{h}(\hat{\mathbf{X}}(t|t-1))) \\ \mathbf{P}(t|t) &= (\mathbf{I} - \mathbf{K}(t) \mathbf{H}_t) \mathbf{P}(t|t-1) \end{aligned}$$

where $\mathbf{h}(\mathbf{X})$ maps the full state to the observable subspace (iGate scan counts, SOR hours, GPS positions), $\mathbf{H}_t = \partial \mathbf{h} / \partial \mathbf{X}|_{\hat{\mathbf{X}}(t|t-1)}$, and $\mathbf{R}(t)$ is measurement noise (GPS accuracy, iGate scan delay, SOR entry latency).

Theorem 8.1 (v1.4 as Marginalized v2.0). The v1.4 Kalman filter is obtained from the v2.0 EKF by:

1. Marginalizing out $\hat{\tau}, W(t), \lambda_b, \rho_b, \mathbf{Q}_c, p_{\text{flow}}, \mathbf{L}, \mathbf{F}$ (all unobserved layers)
2. Treating their contributions as increased process noise $\mathbf{Q}_t^{\text{v1.4}} = \mathbf{Q}_t^{\text{v2.0}}|_{\text{marginalized}}$

The v1.4 prediction errors (unexplained PPH variance) are precisely the variance attributable to the marginalized layers. As each layer becomes observable, $\mathbf{Q}_t$ decreases and prediction improves. This theorem provides a formal quantification of the **value of each new sensor layer** in terms of RMSE reduction.

9. Integrated Optimization Framework

9.1 Real-Time Hub Throughput Maximization

Definition 9.1 (Hub Throughput Objective). The primary operational objective is to maximize total packages sorted by sort close T , subject to sort quality constraints:

$$\max_{\mathbf{u}(\cdot)} \int_0^T \lambda_{\text{sort}}(t) dt \quad \text{subject to:}$$

$$\begin{aligned} \text{SQS}(T) &\geq \text{SQS}_{\min}, \quad \forall b : \rho_b(t) < 1, \quad \forall c : Q_c(t) \leq K_c \\ N_z(t) &\geq N_z^{\min}, \quad s_b(t) \in [s_{\min}, s_{\max}] \end{aligned}$$

where $\mathbf{u}(t) = (\mathbf{N}(t), \mathbf{s}(t), \mathbf{d}(t))$ is the control vector: zone staffing assignments $\mathbf{N}$, belt speed settings $\mathbf{s}$, and door open/close decisions $\mathbf{d}$.

9.2 Pre-Sort Staffing Optimization Under Arrival Uncertainty

Theorem 9.1 (Pre-Sort Optimal Staffing). Given the pre-sort GPS-based volume forecast $\hat{\mathbf{L}}_{\text{final}}$ (Theorem 6.2) and feeder ETA distribution, the optimal initial staffing allocation is:

$$N_z^*(0) = \arg \min_{N_z \geq N_z^{\min}} \mathbb{E} \left[\int_0^T (\lambda_z(t) - N_z(t) \cdot \text{PPH}_z(N_z(t)))^2 dt \right]$$

Under the linear-capacity approximation $\text{PPH}_z(N_z) \approx \overline{\text{PPH}}_z$ (constant per-sorter rate), this reduces to:

$$N_z^*(0) = \left\lceil \frac{\hat{L}_{z,\text{final}}}{T \cdot \overline{\text{PPH}}_z} \right\rceil$$

The uncertainty in $\hat{L}_{z,\text{final}}$ propagates as staffing uncertainty: $\text{Var}[N_z^*(0)] = \text{Var}[\hat{L}_{z,\text{final}}]/(T \cdot \overline{\text{PPH}}_z)^2$, quantifying the staffing risk attributable to GPS ETA uncertainty.

9.3 Belt Speed Control Policy

Theorem 9.2 (Optimal Belt Speed). For a single belt segment with induction rate $\lambda_b(t)$ and cost function penalizing both congestion and over-speed (mis-sort/damage risk):

$$s_b^*(t) = \arg \min_{s \in [s_{\min}, s_{\max}]} [\mathcal{C}_{\text{cong}}(\rho_b(t, s)) + \mathcal{C}_{\text{mis-sort}}(s)]$$

where $\mathcal{C}_{\text{cong}}(\rho) = c_1 \cdot \rho / (1 - \rho)$ (M/G/1 delay cost) and $\mathcal{C}_{\text{mis-sort}}(s) = c_2 \cdot (s / s_{\max})^\eta$ (mis-sort cost increasing superlinearly with speed). The unconstrained optimum is:

$$s_b^* = \left(\frac{c_1 \lambda_b \rho_b}{c_2 \eta / s_{\max}^\eta} \right)^{1/(\eta+1)}$$

projected onto $[s_{\min}, s_{\max}]$. This policy increases belt speed when induction rate is high (prioritizing throughput) and reduces it when packages are irregular or misrouting risk is elevated (prioritizing quality).

9.4 Jam-Conditioned Re-Optimization

Definition 9.2 (Reactive Re-Planning Trigger). A jam event at segment b at time t_{jam} triggers re-optimization of:

1. **Belt speed on adjacent segments** (reduce upstream rate to prevent cascade)
2. **Staffing re-assignment** (redeploy jam-breakers from low-urgency zones)
3. **Door management** (temporarily close high-rate doors feeding b)
4. **Sort completion forecast update** (push end-time estimate with confidence interval)

The re-optimization uses the current EKF state estimate $\hat{\mathbf{X}}(t_{\text{jam}} | t_{\text{jam}})$ as the new initial condition and runs forward simulation to produce updated recommendations within $\delta t_{\text{replan}} < 30$ seconds.

10. Digital Twin Architecture

10.1 What is a Hub Digital Twin?

A **digital twin** of the sort hub is a continuously-updated, high-fidelity simulation of the physical sort that runs in parallel with the live operation, ingesting real sensor data as it arrives and producing:

1. **Current state estimates** — what is happening right now at every belt, chute, and zone
2. **Predictive simulations** — what will happen in the next 15, 30, 60 minutes given current trajectory
3. **Counterfactual analysis** — what would have happened with different control decisions
4. **Prescriptive recommendations** — what should be done now to improve outcome

10.2 Data Ingestion Architecture

Definition 10.1 (Sensor Fusion Hub). The digital twin aggregates data from six stream types:

Stream	Update Rate	Protocol	Latency
iGate scans	Per scan (~100/min peak)	REST API push	<5 sec
SOR entries	Per entry (~1/5 min)	REST API push	<30 sec
GPS telematics	Per vehicle, 10 sec	MQTT stream	<15 sec
Belt speed sensors	Per segment, 1 sec	OPC-UA	<2 sec
Weather API	5-minute intervals	NOAA/NWS REST	<60 sec
Traffic API	1-minute intervals	HERE/Waze REST	<30 sec

All streams are timestamped at source and aligned to a common timeline via the EKF observation model (§8.2), which handles asynchronous observations naturally.

10.3 Predictive Simulation Engine

Definition 10.2 (Monte Carlo Forward Simulation). Given current EKF state estimate $\hat{\mathbf{X}}(t|t)$ and covariance $\mathbf{P}(t|t)$, the digital twin generates $M = 1000$ Monte Carlo trajectories by:

1. Sampling $\mathbf{X}_0^{(m)} \sim \mathcal{N}(\hat{\mathbf{X}}(t|t), \mathbf{P}(t|t))$
2. Propagating each sample forward via the nonlinear dynamics $\mathbf{f}(\cdot)$ with sampled process noise
3. Computing the distribution of any output quantity of interest (PPH at $t+30$, sort completion time, total volume sorted)

The 5th, 50th, and 95th percentile trajectories are displayed as the prediction band — replacing the point estimate currently shown in the v1.4 DOP Calculator with a **probabilistic forecast**.

10.4 Closed-Loop Control Interface

Definition 10.3 (Recommendation Engine). The digital twin’s closed-loop control module runs the integrated optimization (Section 9) at each snapshot and produces a **prioritized recommendation queue** for the coordinator:

```
[18:32] RECOMMENDATION: Redirect 3 sorters from Zone 4 → Zone 7
Reason: Feeder #12 (1,840 pcs, 82% Zone 7) ETA 18:45
Expected PPH gain: +12 PPH in Zone 7
Confidence: 87%
Time window: Act before 18:40

[18:45] ALERT: Belt B-7 density 0.89 (threshold 0.75)
Recommendation: Reduce induction at Door 14 by 30%
Jam probability in next 10 min: 34%
```

Recommendations are ranked by expected throughput impact and confidence, and time-windowed to give the coordinator actionable lead time.

11. Connection to v1.4

11.1 Formal Embedding

Theorem 11.1 (v1.4 as Restricted Case). Every theorem in the v1.4 framework (Theorems 3.1–14.1) remains valid in v2.0. Formally:

$$\mathcal{F}_{v1.4} = \mathcal{F}_{v2.0} \Big|_{\mathbf{x}_{\text{ext}}=\text{unobserved}, \mathbf{x}_{\text{env}}=\text{unobserved}, \mathbf{x}_{\text{infra}}=\text{unobserved}, \mathbf{x}_{\text{pkg}}=\text{unobserved}, \mathbf{x}_{\text{human}}=\text{unobserved}}$$

Proof sketch. v1.4 models only the projection $\mathbf{x}_{\text{ops}}$ of the full state $\mathbf{X}$. All v1.4 quantities (PPH, SQS, $N(t)$, FidelityScore, SQS components) are subcomponents of $\mathbf{x}_{\text{ops}}$. The v1.4 Kalman filter is the Kalman filter for the $\mathbf{x}_{\text{ops}}$ subsystem with inflated process noise absorbing the unobserved layers. The v1.4 Sort Quality Score, jam-breaker distortion corrections, and borrow/loan optimizer are policies defined on the $\mathbf{x}_{\text{ops}}$ subspace — all remain valid restrictions of their v2.0 counterparts. $\square$

11.2 Migration Path: v1.4 $\rightarrow$ v2.0

Each new sensor layer that becomes available incrementally reduces the EKF process noise and improves prediction accuracy. The migration path is:

Step	New Data Source	Expected RMSE Reduction	Complexity Addition
v1.5	SOR real-time push (vs. manual export)	~5–8%	Minimal
v1.6	Belt speed telemetry integration	~8–12%	Low
v1.7	ORION manifest feed (SLIC pre-sort)	~12–18%	Medium
v1.8	Feeder GPS ETA (internal UPS telematics)	~15–25%	Medium
v1.9	P-car GPS / ORION return estimates	~8–12%	Medium
v2.0	Weather + traffic + full EKF	~20–30% total over v1.4	High

Each step along this path is self-contained and independently deployable. v1.5 through v1.9 are the natural incremental extensions of the current validated baseline. v2.0 is the terminus.

12. Data Requirements

For v2.0 to be instantiated in production, the following data streams must exist and be accessible:

12.1 Required Data Sources

Data	Source	Access Mechanism	Status
iGate scan export	Internal iGate API	Current: manual export $\rightarrow$ v1.5: push API	Partially available
SOR hours	Internal SOR system	Current: manual export	Partially available
Feeder GPS positions	UPS Telematics / Coyote	UPS internal API or telematics vendor	Not currently integrated

Data	Source	Access Mechanism	Status
P-car GPS / ORION return	UPS ORION system	UPS internal feed	Not currently integrated
Belt speed sensors	PLC / SCADA system	OPC-UA or existing SCADA interface	Infrastructure exists; API unknown
Induction rate counters	Induction station sensors	Same SCADA	Infrastructure exists; API unknown
Chute load sensors	Load cells (if installed)	SCADA	May not be installed at Chelmsford
Weather data	NOAA NWS / OpenWeatherMap	Public REST API	Easily accessible
Traffic data	HERE / Waze for Cities	Commercial API	Requires subscription
Package dimensions	Vision system (if installed)	Computer vision API or existing QA system	Unknown; may require capital investment
SLIC manifests	UPS Hub Management System	Internal feed	Unknown access level

12.2 Minimum Viable v2.0

A **partial v2.0** achieving the largest prediction improvement at lowest integration cost would require:

1. **Feeder GPS ETA** (largest single gain: ~15–25% RMSE reduction)
2. **NOAA weather API** (free, easily integrated)
3. **Belt speed telemetry** (infrastructure likely exists)
4. **ORION manifest pre-sort feed** (SLIC distribution known before sort start)

These four sources constitute the highest-leverage additions and could be integrated as v1.7–v1.8 milestones before reaching the full v2.0 EKF.

13. Limitations of the Idealized Model

13.1 Sensor Availability and Data Quality

The v2.0 framework assumes all sensor streams are available, reliable, and low-latency. In practice:

- GPS positions update every 10 seconds but may have gaps in dead zones (underground loading docks, thick warehouse walls)
- Belt speed sensors may not exist on all segments at Chelmsford; capital investment may be required

- Weather affects ETAs non-uniformly across the local road network; single-station weather may not capture hyperlocal road conditions
- ORION manifests are probabilistic (drivers may not follow route order); actual return package counts can deviate substantially from forecast

13.2 Model Identifiability

The full v2.0 state vector has \$ 300 dimensions. The iGate/SOR observation vector at each snapshot has \$ 50 dimensions. The system is **severely underidentified** from iGate/SOR alone; the EKF requires meaningful prior distributions on all latent states. In practice, this means:

- Belt density ρ_b must be initialized from physics (belt speed and induction rate, if available) rather than inferred from iGate data
- Fatigue factors $\mathbf{F}(t)$ require either wearable sensor data or strong empirical priors from physiological studies
- Package composition p_{flow} requires either vision systems or historical manifest-based estimates

13.3 Computational Tractability

The EKF Jacobian computation over a 300-dimensional state at 15-minute intervals is feasible ($O(300^2) \approx 90,000$ operations per update step). However, the Monte Carlo simulation engine (1,000 trajectories $\times$ 240 time steps $\times$ state dynamics) requires approximately 2.4×10^8 floating-point operations per forecast — on the order of seconds on modern hardware, acceptable for a 15-minute update cycle, but requiring dedicated compute (not a browser-based tracker).

13.4 Human Behavior Model Limitations

The fatigue and experience models are borrowed from general materials-handling literature. UPS-specific calibration would require:

- Controlled studies linking time-on-task to observed PPH degradation
- Per-sorter experience curves from longitudinal iGate history
- Temperature correlation studies in the Chelmsford facility specifically

These are feasible research projects but require data collection protocols not currently in place.

14. Future Research Directions

14.1 Reinforcement Learning for Real-Time Control

The integrated optimization (Section 9) is solved as a sequence of static problems at each snapshot. A more powerful approach is to formulate the full-sort control problem as a **Markov Decision Process** (MDP):

- **State space:** the v2.0 state vector $\mathbf{X}(t)$
- **Action space:** belt speed adjustments, staffing re-assignments, door open/close decisions
- **Reward:** sort-end throughput minus delay and quality penalties
- **Horizon:** 240 minutes, discretized at 15-minute intervals

Modern deep reinforcement learning (Proximal Policy Optimization, Soft Actor-Critic) could train a policy on simulated sorts generated by the digital twin. The trained policy would then operate

in real-time, mapping observed state to control actions. This represents the full closure of the perception-action loop.

14.2 Multi-Facility Extension

The mathematical abstractions in v2.0 (marked point process, M/G/1 belt queue, fatigue curve, EKF) are facility-agnostic. The parameters (h_0 jam baseline, κ_F fatigue rate, α_w weather attenuation) are Chelmsford-specific. A **federated multi-facility model** would pool data across UPS hubs to improve parameter estimation while allowing facility-specific calibration — a hierarchical Bayesian structure at the facility level, analogous to the per-sorter Beta-Bernoulli model in v1.4.

14.3 Anomaly Detection and Root Cause Analysis

The digital twin provides a principled framework for **anomaly detection**: when observed $\mathbf{z}(t)$ deviates from EKF prediction by more than $k\sigma$, an anomaly is flagged. The EKF residual $\mathbf{z}(t) - \mathbf{h}(\hat{\mathbf{X}}(t|t-1))$ carries causal information: residuals in belt density suggest infrastructure issues, residuals in zone load suggest composition effects, residuals in fidelity suggest scanning compliance issues. Automatic root-cause attribution from the residual pattern is a natural extension.

14.4 Sort Simulation as Training Ground

The digital twin can generate synthetic sorts with controlled parameters (feeder delay, weather event, jam at specific belt, non-flowable surge). These synthetic sorts can serve as **training scenarios** for new coordinators — a formal extension of the LTC system from per-sorter routing training to per-coordinator operational decision training. This closes the three-timescale loop at the management level.

14.5 Jam-Breaker Identification from Belt Telemetry

Currently, jam detection relies on coordinator observation and manual coding. With belt speed sensors and induction rate counters, jam events can be detected automatically (belt speed drops to zero while induction rate remains positive). This would enable automatic application of the v1.4 jam-breaker distortion correction without relying on overhead labor code compliance — reducing the data fidelity requirement that currently limits SQS accuracy.

15. From Files to Equations: A Plain-Language Guide

This section requires no mathematical background. It describes, in plain language, what the software tools in this project actually do, why they were built, and how every decision Rafael makes on the sort floor connects — through those tools — to the mathematical model described in this paper.

15.1 What the Raw Files Actually Are

Every night at the UPS Chelmsford hub, two systems generate data exports:

iGate is the automated package scanning system. Every time a package passes a scanner on the belt, iGate records who scanned it (the employee), what was scanned (the barcode), and when it

happened. At the end of the sort — or at any point during it — a supervisor can pull an export from iGate. This export is a spreadsheet (or CSV) that shows, for each employee, how many packages they scanned during a given time window. From this raw count, you can compute **PPH** (packages per hour): divide total packages scanned by total hours worked, and you have a measure of throughput. This is the heartbeat metric of the sort.

SOR (Staffing and Operations Record) is the manual system where supervisors and coordinators log who worked, in which area, for how long. If a sorter is borrowed from Zone 4 to help Zone 7, that transfer — if logged correctly — appears in SOR. If an employee works overtime, or clocks in late, SOR is where that is recorded. SOR data tells you how many people-hours were actually consumed in each area.

The limitation of both exports is that they are **snapshots in time**. You pull them at 8:30 PM and you get a picture of everything that happened from 6:00 PM to 8:30 PM. Pull them again at 10:00 PM and you get the full sort. If you pull them multiple times during the sort, you get a series of overlapping snapshots — and the **difference** between consecutive snapshots tells you what happened in that interval. This is the core data problem this project was built to solve.

15.2 What the HTML and JavaScript Files Do

The **Hub Operations Tracker** (the `tracker_v1.x.html` file) is a single web page you open in a browser. It has no server — all the logic runs in your browser using JavaScript. Its job is to take those raw iGate and SOR exports and turn them into an operational picture you can actually read during the sort, not just after it.

When you upload an iGate export into the tracker, the JavaScript code reads every row: who scanned, how many packages, over what time window. It then computes PPH for each employee, each area, and the entire hub. It color-codes the results — green means performing well, yellow means marginal, red means struggling — using the same thresholds that the hub’s own dashboard uses, so you are looking at the same interpretation your district managers would see.

When you upload a SOR export, the tracker reads hours by area and employee. It uses this to cross-check the iGate data: are the hours in SOR consistent with the scan records in iGate? If an employee has 300 scans in iGate but SOR shows them working in a different area, the tracker flags this as a fidelity issue. This is the **FidelityScore** in the mathematical model — the fraction of scans that are attributable to employees correctly logged in the right area.

When you upload files from **different times during the same sort** (a 7:45 PM export and a 9:30 PM export, for example), the tracker compares them. It can show you how PPH evolved over the sort, which areas peaked early and which held steady through wind-down, and where things started to slip. This is the **timeline reconstruction** — turning a series of flat spreadsheets into a living operational record.

The **DOP Calculator** tab in the tracker is a projection tool. You tell it: here is my current scan count, here is my current hour count, here is how much time is left. It computes whether you are on track to hit your target volume, and what PPH you need to sustain for the rest of the sort to get there. This is a simplified version of the Kalman filter prediction — it is doing forward projection from the current moment, though without the probabilistic uncertainty bands that the full mathematical model provides.

The **Hub Operations Container** (`hub_operations_v3.x.html`) is a wrapper around the tracker and the operations dashboard — it lets you switch between the live view and the historical view without leaving the browser.

15.3 What the Python Files Do

The **MasterTrendAnalysis extractor** (the Python scripts in the MasterTrendAnalysis folder) does something different from the tracker. Rather than analyzing a single sort in real time, it reads the archive of historical sort exports — 633 of them, spanning multiple years and three distinct operational eras — and structures them into a single dataset.

For each historical sort, the extractor reads the iGate and SOR exports and pulls out the key metrics: total volume, total hours, average PPH, area-level breakdown, date, day of week, season, era (which belt configuration was in use). It saves all of this as a structured JSON file — a clean, machine-readable record of every sort in the archive.

The `MasterTrendAnalysis/app/index.html` file then reads that JSON and lets you visualize trends: how does Chelmsford PPH on a Monday in February compare to a Thursday in November? How did performance change when the second SLS belt was added? The four prediction algorithms (KDE, DTW, weighted median, Kalman filter) run in the browser, each computing its own estimate of “how should today’s sort be going based on all past sorts that looked like today.”

Together, the Python extractor and the HTML prediction app are the **per-sort timescale** of the mathematical framework: they operate on historical data, across sorts, to give context and prediction for any new sort.

15.4 How Rafael’s Decisions Map to the Mathematical Model

Every decision made on the sort floor — every borrow, every loan, every door closure, every belt speed adjustment, every jam response — appears in the mathematical framework as a **control input** $u(t)$.

Here is a plain-language translation of that mapping:

“I’m moving three people from Zone 4 to Zone 7 because Zone 7 is getting buried.”

In the model, this is an update to the staffing vector $N_z(t)$: $N_4(t)$ decreases by 3, $N_7(t)$ increases by 3. The marginal productivity theorem (v1.4 §14) says this move is mathematically optimal if the marginal gain in Zone 7 exceeds the marginal loss in Zone 4 — which is exactly the intuitive judgment you are making when you recognize that Zone 7 is “more buried” than Zone 4 can afford to give up help.

“There’s a jam on B-line. I’m sending two jam-breakers.” This is the jam-breaker distortion model (v1.4 §13). The jam-breakers are physically pulling packages off the belt — real work, real throughput contribution. But because they are coded under an overhead labor code (not a scan-producing area code), iGate records their time as non-productive. The result: denominator (hours) stays high but numerator (scans attributed to them) goes down. The mathematical correction model formalizes this so that future analysis can strip out the artificial PPH suppression their presence caused.

“I’m closing Door 14 because that feeder is stacking up and I can’t induct fast enough.”

This is the induction ceiling theorem (v2.0 §5.2). You are observing $\lambda_{\text{raw}}(t) > C_{\text{induct}}$ — more packages arriving than the belt can take in — and responding by reducing λ_{raw} at that door. The mathematical model tracks the resulting induction buffer B_{induct} and predicts when it will clear, which tells you when you can safely reopen the door without spiking belt density again.

“I’m holding off starting the sort for 10 minutes because the feeder from Boston is almost here.”

This is GPS-based ETA prediction (v2.0 §3). You are implicitly reasoning about the volume distribution over the sort window: starting now with low inbound rate, then getting a 1,800-piece pulse at minute 10, is worse than waiting 10 minutes and having a more uniform inbound rate through the sort. The marked point process model formalizes this reasoning and can compute the expected throughput under both scenarios.

“PPH looks great right now but it’s 8:15 and we’ve barely touched the big Zone 3 feeder.”

This is the phase-aware Markov chain (v1.4 §6.2) combined with volume forecast. You are recognizing that the current phase (Steady, favorable PPH) will give way to a late-arriving volume spike that changes the operational picture. The Kalman filter, when given the feeder ETA, adjusts its sort-end prediction accordingly — what looks like a strong sort at minute 135 may become a compressed wind-down race.

“I’m pushing belt speed up because we’re behind.”

This is the belt speed control policy (v2.0 §9.3). Increasing belt speed raises throughput capacity (μ_b increases) but also raises missort risk (packages pass deflectors faster) and reduces sorter reaction time per package. The optimal policy trades these off explicitly. What you experience as an intuitive judgment — “we’re enough behind that it’s worth the risk” — is the model’s cost function crossing a threshold.

15.5 Why Every Intervention Creates a Ripple

The fundamental insight of the v2.0 framework — and the reason this model is complex — is that the hub is not a collection of independent systems. Every action propagates.

When you move three sorters from Zone 4 to Zone 7, Zone 4’s throughput drops. If Zone 4 was already at its scan-minimum, its chute may now fill more slowly — which sounds fine, but it means downstream trailers load later, which affects outbound sort schedule. Meanwhile Zone 7’s throughput rises, its chute fills faster, and if the loading team hasn’t been notified, you now have full chutes and a bottleneck at the trailer end.

When you close Door 14, the packages staged on the dock don’t disappear. They accumulate. When you reopen the door, they hit the belt in a concentrated burst — a mini-spike that propagates as elevated belt density, increased jam probability, and a short-term PPH dip as sorters process the backlog.

When you send jam-breakers, the zones they came from lose staffing. Their PPH will show a temporary dip in the next iGate snapshot — which may look like a performance problem but is actually a consequence of your correct jam response. The jam-breaker distortion model (v1.4 §13) gives you the mathematical tool to separate “we underperformed” from “we responded to a physical event and the data makes it look like we underperformed.”

In the language of the model: the hub is a **coupled dynamical system**. Every control input $\mathbf{u}(t)$ changes the state $\mathbf{X}(t)$, which then evolves according to the belt queue dynamics, the staffing

model, and the sorter channel — producing new observations at the next iGate snapshot. Your interventions are not corrections to an otherwise stable system. They are participation in the dynamics of the system — each one a perturbation that reshapes the trajectory.

The mathematical framework does not make this more complicated than it already is. It makes it *legible*. The equations give names and structure to what is already happening on the floor. The gap between the current tools (v1.4, the tracker, the extractor) and the idealized model (v2.0, the digital twin) is not a gap in understanding — it is a gap in sensor coverage. The understanding, formalized here, is already complete.

15.6 What This Means for the Tracker Going Forward

The tracker currently tells you what happened. The digital twin tells you what will happen and what you should do about it. The path from one to the other — laid out in the v1.5 through v1.9 migration steps in Section 11.2 — is a series of data integration milestones, each one giving the model more to work with and shrinking the uncertainty in its predictions.

Every time you pull an iGate export mid-sort and upload it to the tracker, you are performing a **Kalman filter update**: you are giving the model a new observation, and the model revises its picture of the current state. Currently, that update is manual — you download, upload, read the color-coded table. In the digital twin, that update happens automatically, every 15 minutes, across all layers simultaneously.

The HTML and JavaScript files are the proof-of-concept. The Python extractor is the data pipeline prototype. The mathematical framework is the blueprint. The digital twin is what they converge toward.

16. CURE Integration: The Outbound Loading Layer

The v2.0 EKF tracks inbound package arrival and sort throughput but terminates at the belt — it models packages being sorted to destination belts but not the downstream loading operation (belt → dock → trailer). CURE provides the missing outbound layer: per-destination-lane trailer utilization, loading density, and throughput rates.

Definition 16.1 (Outbound Loading State). For destination lane d at time t , the outbound loading state is:

$$\mathbf{o}_d(t) = (U_d(t), \text{PPW}_d, N_d^{\text{wall}}(t), \lambda_d^{\text{load}}(t))$$

where $U_d \in [0, 1]$ is trailer utilization, PPW_d is pieces per wall position (loading density), N_d^{wall} is number of active wall positions, and λ_d^{load} is the loading rate (packages/minute).

The CURE data provides $U_d(t_{\text{close}})$ — the final trailer utilization at sort close — and historical averages of PPW_d and piece counts. The v2.1 extension models $\lambda_d^{\text{load}}(t)$ as a derived quantity from the induction rate:

$$\lambda_d^{\text{load}}(t) = \eta_d \cdot \lambda_d^{\text{sort}}(t)$$

where $\lambda_d^{\text{sort}}(t)$ is the rate at which packages are sorted to belt d (from the v2.0 induction model) and $\eta_d \in [0, 1]$ is the loading efficiency for destination d .

Chronic high-utilization lanes. Across all four sorts with CURE data (05.05–05.08), LL BEAN (destination 0432) appears at utilization 88–98%. This represents a structural loading constraint: Chelmsford consistently produces volume at or near LL BEAN trailer capacity. When $U_{\text{LL BEAN}} > 0.95$, the loading dock becomes the binding constraint — additional sort throughput on the LL BEAN belt does not improve service; it creates a dock queue.

CURE Outbound Utilization — Five-Sort-Week Summary:

Date	Active Lanes	LL BEAN Util%	Top Congested Lanes
05.05 Tue	143	98.4%	LL BEAN, Stratham (80.6%), Columbus (80.2%)
05.06 Wed	109	94.5%	LL BEAN, Willow Grove (88.8%), Saugus (82.9%)
05.07 Thu	106	95.7%	LL BEAN, Bell Hub (87.7%), Willow Grove (86.7%)
05.08 Fri	102	88.9%	LL BEAN, Hartford (83.1%), St. Johnsbury (82.5%)

Theorem 16.1 (Loading Bottleneck Regime). For a destination lane d with $U_d(t) > U_d^{\text{cap}} \approx 0.92$, the marginal utility of additional sort throughput λ_d^{sort} is:

$$\frac{\partial \text{SQS}}{\partial \lambda_d^{\text{sort}}} < 0$$

because the additional packages join a dock queue, increasing the probability of HUB DNS RNP events (driver departure before loading completes) rather than improving sort throughput.

Corollary. The coordinator’s optimal action when CURE shows a lane above U_d^{cap} is **dock-side intervention** (adding loaders, requesting loading support, or staging packages to a secondary location) rather than increasing belt speed or sorter count for that zone.

CURE EKF Integration. The v2.0 EKF state vector $\mathbf{X}(t)$ is augmented with the outbound loading state:

$$\mathbf{X}_{v2.1}(t) = \begin{bmatrix} \mathbf{X}_{v2.0}(t) \\ \{U_d(t)\}_{d \in \mathcal{D}} \\ \{\lambda_d^{\text{load}}(t)\}_{d \in \mathcal{D}} \end{bmatrix}$$

where $\mathcal{D}$ is the set of active destination lanes (102–143 per sort, decreasing with hub volume). The CURE export (available $t+24\text{h}$) provides the ground-truth $U_d(t_{\text{close}})$ for the EKF correction step.

CURE as volume-gradient predictor. The lane count decline (143→102 Tue→Fri) is a direct consequence of the volume gradient (113K→90K packages). The relationship is approximately linear: each 10K package reduction eliminates ~13 active destination lanes. This gives the EKF a structural prior for Friday-night lane count given the week’s volume trajectory.

17. SEAS Package Composition: Physical State Extension

The v2.0 model includes a package composition layer (§6: flowable fraction α_f , irregular fraction). SEAS provides the empirical calibration data for these parameters at the belt level.

SEAS Package Composition Variables: - Smalls (Not Bagged): Packages classified as small that were not placed in a smalls bag — these are the flowable packages handled by the Smalls Sort area. - **Bags Created / Bags Linked:** Smalls bags created and linked to destination — unlinked bags become BAG NOT LINKED MISLOAD events. - **Scan% Efficiency:** Fraction of packages that were scanned vs. reported (physical count).

05.04 Ground Truth (from SEAS_Twilight): - Hub total SEAS scans: 122,856 - SMALLS SORT: 47,501 scans — the primary smalls handling area, producing ~40% of all hub quality events - Hub scan efficiency: 98.7% - Total misloads: 78 (SMALLS SORT=29, PD-11=9, PD-6=4, PD-3=4, PD-9=3, UNASSIGNED=18) - Total LIB: 153 (UNASSIGNED=99, SMALLS SORT=42, PD-7=6, PD-6=3) - UNASSIGNED row = SOR mismatch proxy (packages scanned without area assignment)

Proposition 17.1 (Smalls LIB Concentration). Smalls Sort generates a disproportionate fraction of hub LIB events relative to its volume. On 05.04, SMALLS SORT produced 42 of 153 LIB events (27.5%) from 47,501 of 122,856 hub scans (38.7% of scans). Smalls sort quality is the highest-leverage quality intervention point in the hub.

Basis: The smalls bagging chain has two additional failure modes vs. regular packages: (1) bag creation (bag not created = package left unbagged and mis-sorted), (2) bag linking (bag created but not linked to correct destination = BAG NOT LINKED MISLOAD). Regular packages have only one failure mode: physical mis-sort. The additional failure chain stages elevate the smalls LIB rate vs. regular package LIB rate.

SEAS as EKF observable. The v2.0 EKF's observation vector $\mathbf{y}(t)$ is extended with SEAS quality metrics (available $t+24h$ with the next-day cadence):

$$\mathbf{y}_{v2.1}(t) = \begin{bmatrix} \mathbf{y}_{v2.0}(t) \\ \text{Misload}_{\text{hub}}(t) \\ \text{LIB}_{\text{hub-derived}}(t) \\ \text{Smalls}_{\text{unbagged}}(t) \\ U_d(t_{\text{close}}) \text{ for } d \in \mathcal{D} \end{bmatrix}$$

The SEAS metrics arrive with a 24-hour lag, so they feed the correction step for the *previous* sort's EKF trajectory, not the current sort's real-time state.

18. LIB as Service Failure State: EKF Integration

LIB events represent the operational manifestation of the emergence gap: when the hub's collective output ($\text{PPH} \times \text{sorters} \times \text{time}$) falls short of the service requirement (all packages processed before sort close), LIB events accumulate. They are the observable consequence of the purposeful systems dynamics formalized in the v3.x series.

Definition 18.1 (Service Failure Rate). For sort s , the service failure rate is:

$$\text{SFR}(s) = \frac{\text{LIB}_{\text{HUB DERIVED}}(s)}{V_{\text{PD}}(s)} \cdot 10^4$$

(expressed per 10,000 packages to avoid small decimals)

Five-Sort SFR and LIB Breakdown:

Sort	HUB DERIVED	DNS RNP	REPORTED	OPL	CENTER	CENTER LIB	Total LIB	SFR (per
05.01	125	15	16	5	8	1	170	~21.4
05.04	99	28	13	4	9	1	154	8.4
05.05	75	26	55	5	6	1	168	6.6
05.06	67	27	21	5	4	1	125	6.0

Note: 05.01 SFR is approximately $3.5\times$ higher than 05.04–05.06. This is a genuine anomaly — either the May 1 sort had a structural issue, or the SEAS data covers a different effective volume than the iGate data. The SFR metric should be computed consistently from SEAS scans (not iGate gross volume) for valid cross-sort comparison.

Theorem 18.1 (LIB Decay Property). Under constant staffing and volume, HUB DERIVED LIB count is monotone decreasing over a working week, approaching a floor set by the structural LIB rate — packages that cannot be processed in any 4-hour window due to arrival timing, regardless of throughput. The structural floor is estimated at approximately 50–60 HUB DERIVED LIB events for a Chelmsford Twilight sort at 100K+ volume, based on the observed minimum across the five-sort week (67 on 05.06 with the week still ongoing at time of writing).

EKF Correction. The LIB count at sort close provides a correction signal for the EKF’s throughput estimate. If the EKF predicted $\hat{V}(t_{\text{close}})$ packages processed but $\text{LIB}_{\text{derived}} > 60$ events, the EKF has over-estimated effective throughput — the hidden state (zone PPH $\times$ hours) was lower than modeled. Formally:

$$\Delta \hat{V}_{\text{EKF}} = -\mathbf{K}_{\text{LIB}} \cdot (\text{LIB}_{\text{derived}}(s) - \text{LIB}_{\text{floor}})$$

where $\mathbf{K}_{\text{LIB}}$ is the Kalman gain for the LIB observable and $\text{LIB}_{\text{floor}} \approx 55$ is the structural floor.

19. Zone 9-12 Structural Constant: Empirical Calibration

The five-sort dataset reveals a structural constant of the Chelmsford hub’s destination mix that is invisible in any single sort’s data.

Hub PD Belt Volumes — Five-Sort Week (iGate Gross Volume, final snapshot):

													Z9- Hub PD
Date	PD01	PD02	PD03	PD04	PD05	PD06	PD07	PD08	PD09	PD10	PD11	PD12	12 Total
05.04 Mon	9,391	9,396	8,508	12,076	12,985	9,860	9,821	7,555	9,595	6,160	10,500	12,292	38,547
													118,139

Date	PD01	PD02	PD03	PD04	PD05	PD06	PD07	PD08	PD09	PD10	PD11	PD12	Z9-12	Hub PD Total
05.05 Tue	11,502	7,853	8,547	10,338	14,178	8,235	9,938	6,972	9,035	7,167	9,687	9,816	35,705	113,268
05.06 Wed	11,414	8,402	8,154	9,450	13,634	8,690	9,696	7,707	9,482	6,897	8,820	8,803	34,002	111,149
05.07 Thu	11,498	7,747	7,641	7,911	11,056	7,003	8,460	7,595	7,845	6,101	8,555	7,304	29,805	98,716
05.08 Fri	9,339	6,245	5,813	6,027	11,333	8,484	8,696	6,598	7,315	5,817	7,610	6,599	27,341	89,876

$Z9-12 = PD09HVD + PD10HVD + PD11HVD + PD12HVD$ (all four HVD-suffixed belts). *iGate Gross Volume* is the authoritative volume figure.

Theorem 19.1 (Zone 9-12 Invariance). Zone 9-12 (belts PD09-PD12) maintains a structurally constant share of total hub PD volume across the weekly volume gradient, within ± 2 percentage points:

$$\frac{V_{Z9-12}(s)}{V_{PD}(s)} = 0.311 \pm 0.013 \quad \forall s \in \{05.04, 05.05, 05.06, 05.07, 05.08\}$$

Empirical basis: 32.6%, 31.5%, 30.6%, 30.2%, 30.4% across five sorts. Mean = 31.1%, standard deviation = 0.9 percentage points.

Operational consequence for the EKF. The EKF prior for V_{Z9-12} can be set as:

$$V_{Z9-12} \sim \mathcal{N}(0.311 \cdot V_{PD}^{\text{forecast}}, (0.013 \cdot V_{PD}^{\text{forecast}})^2)$$

This gives the coordinator an immediate prior on Zone 9-12 volume before the sort begins, conditional only on the hub-level volume forecast (available from the SOR planned volume field).

Belt-level consistency within Zone 9-12. Across the five sorts, PD09 and PD12 are consistently the highest-volume belts within the zone (PD09: 7.3–9.6K, PD12: 6.6–12.3K) while PD10 is consistently the lowest (5.8–7.2K). This suggests PD10’s destination mix is thinner (fewer addresses in that sort group), which has implications for sorter-to-belt assignment optimization.

20. Weekly Volume Gradient: Structural Model

Definition 20.1 (Twilight Volume Envelope). For the Chelmsford Twilight sort, the weekly volume envelope $V(d)$ for day-of-week $d \in \{Mon, Tue, Wed, Thu, Fri\}$ follows:

$$V(d) = V_{\text{Mon}} \cdot \gamma^{d-1}$$

where $\gamma \approx 0.938$ is the day-of-week decay factor estimated from the five-sort week (118,139 $\rightarrow$ 89,876 over 4 days: $\gamma = (89876/118139)^{1/4} = 0.938$).

Model Fit vs. Actual (05.04–05.08):

Day	Model Forecast	Actual	Deviation
Monday	118,139 (anchor)	118,139	0.0%
Tuesday	110,898	113,268	+2.1%
Wednesday	104,023	111,149	+6.5% (anomaly)
Thursday	97,574	98,716	+1.2%
Friday	91,527	89,876	−1.8%

The Wednesday deviation (+6.5%) is the week’s primary anomaly — actual volume exceeded the geometric model. This is consistent with the Wednesday SOR staffing anomaly documented in the v3.6 parallel research branch (planned 120 workers vs. actual 236), which created excess induction capacity that drew in volume that otherwise might have been deferred.

Theorem 20.1 (Gradient-Invariant Zone Share). Despite the 24% Monday-to-Friday volume decline, the proportional distribution across zones (specifically Zone 9-12 invariance from Theorem 19.1) implies that the EKF’s zone-level volume forecast degenerates to a scalar multiplication of the hub-level forecast. The coordinator does not need zone-specific volume priors — they need only the hub-level forecast and the structural share constants:

$$\hat{V}_z(s) = \phi_z \cdot \hat{V}_{PD}(s) \quad \text{where } \phi_{Z9-12} = 0.311$$

This dramatically simplifies the pre-sort planning computation from a 12-dimensional zone forecast to a 1-dimensional hub forecast scaled by fixed zone shares.

21. Updated v2.0 Theorem Status Under Empirical Testing

This section reports the empirical status of the major v2.0 theorems against the five-sort week data.

Theorem 3.1 (GPS-based volume arrival forecast): *Partially supported.* The volume gradient model (§20) provides a structural prior without GPS data, suggesting that the GPS-based forecast is most valuable for same-day variance (feeder-specific delay) rather than day-level forecast, which is well-captured by the day-of-week envelope. GPS remains essential for within-sort timing but adds less to the weekly level forecast than originally hypothesized.

Theorem 6.1 (Effective PPH as function of flowable/non-flowable mix): *Supported.* SEAS confirms that SMALLS SORT is a distinct throughput regime from regular PD sorting, with its own quality failure chain. The flowable fraction α_f should be estimated from SEAS smalls vs. non-smalls volume rather than package dimension data, which is not currently available.

Theorem 8.1 (v1.4 as marginal of v2.0): *Supported and extended.* The LIB events and CURE utilization data are exactly the unobserved layers that v1.4 marginalizes out. Prediction errors in v1.4’s PPH model correspond to variance in {LIB rate, CURE utilization, SEAS smalls composition} — the layers now observable in v2.1.

Theorems 9.1/9.2 (Optimal belt speed control): *Require amendment at loading bottleneck lanes (Theorem 16.1 corollary).* At LL BEAN-class lanes (chronic $U_d > 0.92$), increasing belt speed does not improve service quality. The optimal control law must condition on CURE utilization:

$$\mathbf{u}^{\text{opt}} = \begin{cases} \mathbf{u}_{v2.0}^{\text{opt}} & \text{if } U_d < 0.92 \\ \text{dock-side intervention} & \text{if } U_d \geq 0.92 \end{cases}$$

Data quality note. All volume figures in §§19–20 use iGate Gross Volume as the authoritative source. SOR volume is used only for organizational context, not ground-truth volume calculations. SEAS test employee IDs (0000000001, etc.) and LabelTrainingCertification test IDs (1111111, 2222222) are excluded from all quality metrics.

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Appendix A: Session Log

Session	Date	Framework Version	Focus
1–3	2025	v1.0–v1.1	Routing, channel model, monoid algebra, corrections
4	2026-05	v1.2	KDE, DTW, Kalman, MLR, CUSUM research
5	2026-05-12	v1.3	Academic restructure, 13 theorems, 35+ citations
6	2026-05-12	v1.4	SQS, Jam-Breaker distortion, Predictive staffing
7	2026-05-12	v2.0	Idealized digital twin: GPS, weather, belt physics, EKF

Session	Date	Framework Version	Focus
8	2026-05-12	v2.1	§16: CURE outbound loading layer — Definition 16.1 (outbound loading state); Theorem 16.1 (loading bottleneck regime: SQS/ $_sort$ < 0 above $U_cap\ 0.92$); CURE EKF augmentation; lane count as volume- gradient proxy (143→102 lanes, Tue→Fri). §17: SEAS package composition — smalls LIB concentration (Proposition 17.1: 27.5% of LIB from 38.7% of scans); SEAS as t+24h EKF correction observable. §18: LIB as service failure state — Definition 18.1 (SFR per 10K pkgs: 8.4 on 05.04, 6.0 on 05.06); Theorem 18.1 (LIB decay property; structural floor ~50–60 derived LIB); EKF correction via LIB at sort

Session	Date	Framework Version	Focus
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Appendix B: New Symbol Glossary (v2.0 Additions)

Symbol	Definition
Φ	Inbound freight marked point process
τ_i	Arrival time of vehicle i
$\mathbf{m}_i$	Mark of vehicle i : $(V_i, \mathbf{s}_i, \text{type}_i)$
V_i	Piece count for vehicle i
$\mathbf{s}_i$	SLIC load vector for vehicle i
$\hat{\tau}_i(t)$	GPS-based ETA estimate at time t
$\sigma_{\tau,i}$	ETA uncertainty (std dev, minutes)
$W(t)$	Weather state at time t
α_w	Weather speed attenuation factor
λ_{acc}	Local accident rate per unit road length
γ_w	Weather accident multiplier
$\lambda_b(t)$	Package arrival rate at belt segment b
$\mu_b(s_b)$	Belt service rate at speed s_b
$\rho_b(t)$	Belt occupancy (traffic intensity)
$h_b(t)$	Jam hazard rate at segment b
$Q_c(t)$	Chute queue length at chute c
K_c	Chute physical capacity
$p_{\text{flow}}(t)$	Flowable package fraction
$\mathbf{L}(t)$	Zone load vector
$F_j(t)$	Fatigue factor for sorter j
κ_F	Fatigue rate parameter
γ_F	Fatigue onset exponent
e_j	Sorter experience (sorts completed)
$\mathbf{X}(t)$	Full v2.0 state vector
$\mathbf{F}_t$	EKF Jacobian of dynamics
$\mathbf{H}_t$	EKF Jacobian of observation
$\mathbf{K}(t)$	Kalman gain matrix
$\mathbf{Q}_t$	Process noise covariance
$\mathbf{R}(t)$	Measurement noise covariance

All v1.4 symbols remain unchanged and carry over to v2.0 and v2.1.

v2.1 Additional Symbols:

Symbol	Definition
$\mathbf{o}_d(t)$	Outbound loading state for destination lane d
$U_d(t)$	Trailer utilization for lane d at time t [0,1]
PPW_d	Pieces per wall position (loading density) for lane d

Symbol	Definition
$N_d^{\text{wall}}(t)$	Number of active wall positions for lane d
$\lambda_d^{\text{load}}(t)$	Loading rate for lane d (packages/minute)
η_d	Loading efficiency for destination d [0,1]
U_d^{cap}	Loading capacity threshold (0.92) above which bottleneck regime applies
$\mathcal{D}$	Set of active destination lanes (102–143 per sort)
$\text{SFR}(s)$	Service failure rate for sort s (HUB DERIVED LIB per 10K packages)
$\text{LIB}_{\text{floor}}$	Structural LIB floor (55 HUB DERIVED events per Twilight sort)
ϕ_z	Structural zone share constant ($\phi_{\text{Z9-12}} = 0.311$)
γ	Day-of-week volume decay factor (0.938 for Chelmsford Twilight)
$V_{\text{Z9-12}}(s)$	Zone 9-12 aggregate volume for sort s

Version: 2.1 — Empirical Grounding Extension (Five-Sort-Week Data, 05.04–05.08 2026)

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Facility: UPS Chelmsford Hub

Branch: Research vision — extends v2.0 with empirical calibration; separate from v1.4 production baseline and v1.5 incremental path

Date: May 2026

The v1.4 framework captures what we can measure. The v2.0 framework describes what we would know if we could measure everything. The v2.1 framework measures what we actually found when five sorts of real data arrived.